

MIT Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)		ASSIGNMENT(S)	
Alandi (D), Pune - 412105		ACADEMIC YEAR	2025-26
SCHOOL OF COMPUTER ENGINEERING		SEMESTER	VI
		CLASS	TY BTech
		DIVISION	A

COURSE CODE	2304314T	ASSIGNMENT NO	1
COURSE	DESIGN AND ANALYSIS OF ALGORITHMS	DATE	3/27/2026

QUE. NO.	QUESTION DESCRIPTION [with marks distribution]	RBT Level	CO NO.	Marks
1	List the following functions according to their order of growth from the lowest to the highest : n^2 , n^2-n , n^3 , n^3+234 , n^2+99 , n . Also justify your answer using Big-O notation.	Level 2	1	5
2	Find the Huffman encoding of your full name (first, middle, last) with the conditions ignore spaces, convert all letters to lowercase, consider only alphabets (a–z) Calculate total transmission cost (in bits).	Level 3	2	7
3	Consider 4 elements $a1 < a2 < a3 < a4$ with: $p1 = 1/4$, $p2 = 1/8$, $p3 = 1/16$, $p4 = 1/16$ $q0 = 1/4$, $q1 = 3/16$, $q2 = q3 = q4 = 1/16$. a) Construct the Optimal Binary Search Tree as a minimum cost tree. b) construct the table of values W_{ij} , C_{ij} , and R_{ij} , computed by the algorithm to compute the roots of optimal sub-trees.	Level 3	3	8

Rubrics

Criteria	CO Addressed	Exceptional Quality	Best Quality	Average Quality	Low Quality	Lowest Quality
		(5/7/8 Points)	(4/6/7 Points)	(2/4/5 Points)	(1/2/4 Point)	(0 Point)
Question 1	CO1	Correctly identifies growth rates using Big-O notation	Final order is correct Minor issues: Missing explanation OR Slight confusion in grouping equivalent terms	Significant errors in ordering Does not clearly apply dominant term idea Only 1–2 positions correct	Very little understanding Random or mostly incorrect ordering No explanation	No attempt made OR Answer is completely incorrect / irrelevant No understanding of Asymptotic Analysis
Question 2	CO2	Correctly lists all characters in the full name (first, middle, last) Computes accurate frequency of each character Constructs the Huffman tree correctly Generates valid prefix codes for each character Correctly encodes the full name Calculates total transmission cost (bits) accurately Work is clearly shown step-by-step	Minor mistakes in: Frequency count OR Code assignment (small errors) Huffman tree mostly correct Encoding mostly correct Transmission cost slightly off due to minor errors Good understanding shown	Partial attempt: Frequency table incomplete or incorrect Huffman tree incorrectly formed Codes may violate prefix property OR are inconsistent Transmission cost incorrect or missing Limited understanding of the algorithm	Very minimal work: Only lists characters OR partial frequencies No proper tree construction No valid encoding No transmission cost	No attempt OR completely irrelevant answer No understanding of Huffman coding process

Question 3	CO3	Correctly applies Optimal Binary Search Tree method Accurately computes all: wij (weights) Cij (costs) rij (roots) Constructs complete tables without errors Correctly builds the optimal BST structure Final tree has minimum expected search cost Clear, well-organized presentation	Minor calculation mistakes in tables (1–2 small errors) Overall method and approach are correct Optimal BST structure is mostly correct Tables wij,Cij,rij are mostly complete Logical steps are shown clearly	Understands the algorithm but: Several calculation errors in tables Some missing entries in wij,Cij,rij BST constructed but not fully optimal Steps are partially correct but inconsistent	Partial attempt: Only some wij values computed correctly Cost/root tables mostly incorrect or incomplete BST construction is incorrect or missing key structure Shows limited understanding of dynamic programming approach	No attempt or completely irrelevant answer
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COURSE TEACHER

Name and Sign with date